

Victorine Maikem

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Professional Summary

Graduate student in Health Informatics at the University of Toronto with hands-on experience in machine learning, AI software development, and applied digital health research. Skilled in developing and deploying ML models, managing messy real-world datasets, and building AI-driven applications. Experienced with TensorFlow, PyTorch, Keras, Scikit-Learn, and cloud-based deployment (AWS, Docker, Kubernetes). Passionate about applying AI to real-world healthcare and industry problems, and contributing to Canada's AI leadership through open-source development and collaborative research.

Technical Skills

- **Programming & ML:** Python, TensorFlow, PyTorch, Keras, Scikit-Learn, NumPy, Pandas, SQL, JavaScript
 - **Deep Learning & AI:** Model training, hyperparameter tuning, architecture optimisation, CNNs, predictive modelling
 - **Data Engineering:** ETL pipelines, unstructured & structured data processing, real-world messy datasets
 - **Software Development:** Flask, Django, REST APIs, Docker, Kubernetes, Git, Agile methodologies
 - **Cloud & DevOps:** AWS (EC2, S3, Lambda), Terraform, CI/CD pipelines
 - **Visualisation & Tools:** Jupyter Notebook, Tableau, Apache Spark, Streamlit
 - **Collaboration:** Agile teamwork, stakeholder engagement, knowledge translation
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Professional Experience

AI4PH Internship Lead,

Artificial Intelligence for Public Health, University of Toronto, Dalla Lana School of Public Health | Nov 2025 – Present

- Supporting the co-development of internship projects with partner organisations by aligning research priorities and defining clear deliverables.
- Mapping the user journey for applicants, interns, and alumni to inform process improvements and strengthen engagement.
- Assisting in the design and coordination of the internship application process, including call preparation, screening, and shortlisting.
- Contributing to program delivery and tracking systems using MS Office, Slack, and OneDrive to enhance operational efficiency and communication.

Software Developer

Midark, Remote | Jun 2023 – Present

- Designed and deployed 6+ commercial websites, improving visibility by up to 30%.
- Developed Teachera, a full-stack engagement platform (React/Django/Postgres) that reduced admin workload by 40% and increased engagement by 65%.
- Consistent updating and maintaining of all platforms, reducing downtime.

Health Data Science Intern

Africa Centres for Disease Control and Prevention (Africa CDC), Science & Innovation Directorate | May 2025 – Aug 2025

- Developed and deployed a continental research capacity platform (Python/Streamlit) mapping 150+ research sites across six West African countries, supporting strategic planning and investment decisions.
- Engineered ETL pipelines and REDCap data systems, improving data quality, harmonising multi-source datasets, and creating standardised data dictionaries aligned with the Africa CDC Member States Data Sharing Agreement.
- Designed analytical dashboards integrating clinical, epidemiological, socio-ecological, and pharmacovigilance datasets to support research site mapping, Mpox outbreak monitoring, and evidence-based research prioritisation.
- Improved functionality and supported maintenance of Africa CDC's research prioritisation web application, enhancing usability and performance.
- Produced policy briefs and knowledge-translation infographics, strengthening communication with ministries of health.

Database Developer Intern

Vanguard Economics LTD, Kigali, Rwanda | Jun 2023 – Dec 2023

- Engineered and optimised MySQL schemas, reducing query latency by 40%.
- Built Django backend services and secure RESTful APIs, improving system integration and performance.

Web Developer Intern

Midori Agro Ventures, Lusaka, Zambia | May 2023 – Oct 2023

- Developed a custom web platform, improving client retention by 10%.
- Automated workflows, boosting engagement by 20%.

Project Intern

Ecoplanet Bamboo, Kigali, Rwanda | May 2022 – Aug 2022

- Built a real-time data visualisation dashboard (Python/Streamlit), cutting manual reporting by 60%.
- Conducted financial data reconciliation, enhancing forecasting accuracy.

Research & Collaboration Experience

Volunteer Research Collaborator – AI in Action: Transforming Clinical Care Across Canada

Digital Health Canada | Jan 2025 – Present

- Analysed AI-driven clinical solutions (diagnostics, workflow automation) across provinces.
- Supported the design of a Clinical AI Readiness Toolkit and deployment map to guide healthcare adoption.
- Engaged clinicians and researchers to identify barriers and best practices in AI adoption.

Additional Research Contributions

- Contributed to a Systematic Review and Meta-Analysis on clinical presentation, transmissibility, and virulence of Mpox Clades 1a, 1b, and 2b (including G.1) under the Africa CDC Science & Innovation Directorate.
- Authored academic commentaries on AI Ethics Training in Low- and Middle-Income Countries (Joseph et al., 2025) and AI Adoption in Genomics Across Africa (Tessema et al., 2025), supporting Africa CDC's thought leadership in responsible and equitable AI.

Key Projects

Counterfeit Drug Detection System (Capstone) | Apr 2024

- Designed and trained deep learning models (TensorFlow/Keras, PyTorch prototyping), achieving 80%+ accuracy.
- Conducted hyperparameter tuning and architecture refinement to improve model performance.
- Built a Flask web interface for real-time deployment and user interaction.

Movie Prediction System | Apr 2023

- Built predictive ML models (regression, clustering, collaborative filtering) for box office forecasting.
- Designed preprocessing pipelines to enhance accuracy.

Water Management System | Nov 2021

- Developed a Python-based monitoring system integrating real-time sensor data and predictive analytics.
- Reduced household water waste by 35%.

Leadership & Volunteer Experience

- Conference Volunteer, Canadian Pain Society Annual Conference (2025)
- Fellow, One Million Leaders Africa (2023–Present)
- Chairperson, ALU Christian Fellowship (2022–2023)
- Microsoft Learn Student Ambassador (2022–Present)
- Volunteer, Zoe Health Foundation & Welisane Foundation (2020–Present)

Certifications

- Google & Kaggle Gen AI Intensive Course (2025)
 - Stanford Online Data science in precision medicine (2025)
 - HELINA HL7 FHIR Certification (2025)
 - Udacity Nanodegree, Full Stack Development (2022)
 - Africa Agility Bootcamp, Data Science & Machine Learning (2022)
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